Dietary antioxidant intake, allergic sensitization and allergic diseases in young children.

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BACKGROUND: Allergic diseases have risen in prevalence over recent decades. The aetiology remains unclear but is likely to be a result of changing lifestyle and/or environment. A reduction in antioxidant intake, consequent to reduced intake of fresh fruits and vegetables, has been suggested as a possible cause. OBJECTIVE: To investigate whether dietary antioxidant intake at age 5 was related to atopy at 5 and 8 years of age amongst children in an unselected birth cohort. METHODS: Children were followed from birth. Parents completed a validated respiratory questionnaire and children were skin prick tested at 5 and 8 years of age. Serum IgE levels were measured at age 5. At age 5, antioxidant intake was assessed using a semi-quantitative food frequency questionnaire (FFQ). A nutrient analysis program computed nutrient intake, and frequency counts of foods high in the antioxidant vitamins A, C and E were assessed. RESULTS: Eight hundred and sixty-one children completed both the respiratory and FFQ. Beta-carotene intake was associated with reduced risk of allergic sensitization at age 5 [0.80 (0.68-0.93)] and 8 [0.81 (0.70-0.94)]. In addition, beta-carotene intake was negatively associated with total IgE levels ($P = 0.002$). Vitamin E intake was associated with an increased risk of allergic sensitization [1.19 (1.02-1.39)], only at age 5. There was no association between antioxidant intakes and wheeze or eczema. CONCLUSION: Increased beta-carotene intake was associated with a reduced risk of allergic sensitization and lower IgE levels, in 5- and 8-year-old children. Dietary antioxidants may play a role in the development of allergic sensitization.